

Team Description

Summary of Technical Strengths

Ben T: Ben is a computer science major. His background in software development includes Python, C, and C++, along with some web development experience using Vue, Typescript, and Firebase. He also has experience with Excel, SQL, and Git, along with a strong interest in data analysis and statistics. This past summer, he interned at Orka Automation, where he worked on the company's website and used Excel to analyze inventory data and sales projections.

Wyatt W: Wyatt is a senior majoring in computer science. His background in software development includes web development using Vue and Google Firebase, as well as experience with game programming and design. His programming strengths include C, microcontroller programming, and developing interactive applications. He also has a strong background in mathematics, including differential equations and calculus-based physics.

Sam R: Sam is a senior who is majoring in computer science. His experience in programming includes Java, Python, and C, as well as some slight familiarity with Javascript. He has experience in creating UI elements using Pygame to ensure proper user friendliness. He has worked with SQL and Git, and has a deep interest in math.

Jason B: Jason is a senior majoring in computer science and minoring in cybersecurity and mathematics. He has experience with Python, Java, C, and C++. He has worked with SQL and Git. Jason also has experience with game programming and has worked with Unity and Godot.

Anticipation of Growth Areas

Ben T: Ben has some experience with Vue, TypeScript, and Firebase from a previous class, but plans to build on that by working with Spherical Easel's larger, more complex codebase. Since the project centers on guided tours for new users, he also wants to grow his UI/UX skills, and research how similar tour systems like Desmos are built.

Wyatt W: Wyatt has experience with web development using Vue and Google Firebase, but has limited experience working on a large existing web application and navigating an established codebase. Through this project, he hopes to gain more experience developing within a larger software system and working with TypeScript. He also wants to improve his skills in user interface and user experience design, particularly when creating intuitive and interactive tutorials for new users. Additionally, working on Spherical Easel will provide experience collaborating with a team using Scrum, GitHub, and professional software development practices.

Sam R: Sam has some experience with Javascript via self-learning and a brief overview of it in a programming languages course. He has worked on making UI, but that was in Python using the Pygame module. However, he has not had the opportunity to work with Vue, Typescript, or Firebase, but with a keen interest in web development, he hopes to be able to build and grow these skills to ensure client satisfaction.

Jason B: Jason doesn't have much experience making UI. He has never used Vue or TypeScript, and he has very little experience using Firebase. Jason is hoping to grow these skills. He also has never done a project for a client before, so he is hoping to grow his communication skills.

Project Description

Background

Online tools for in the classroom learning for math have come a long way. One of the most notable platforms that includes such tools is Desmos, which provides graphing and scientific calculators. These tools come with online tutorials that help guide new users on how to access specific features within their platform. Spherical Easel takes the concept of spherical geometry and puts it to the test. This tool allows students/researchers to facilitate the learning (and teaching) of spherical geometry. With a tool as vast as Spherical Easel there are many different options and features that allow for many different use cases. This can be very daunting for new users of the website. Our main focus on this project is to be able to create interactive tours/tutorials that will help guide users on how to use the platform. Our goal is to make it as intuitive as possible, making sure that the user can get comfortable using the web app.

Description of Intended Features/Backlog

1. Add a Point Tour that guides the user through activating the Point tool and clicking on the main sphere canvas to create a point on the front of the sphere.
2. Rotate Sphere Tour that guides the user through creating a geometric object, activating the Rotate Sphere tool, and clicking and dragging on the sphere canvas to rotate the sphere, including rotation with momentum.
3. Add a Segment Tour that guides the user through activating the Segment tool and selecting two locations on the sphere canvas to create a line segment, with an alternate method demonstrating creation by clicking and dragging.
4. Add a Point to the Back of the Sphere Tour that guides the user through activating the Point tool and holding the Shift key while clicking on the sphere canvas to create a point on the back of the sphere.
5. Create a Line Segment Longer Than Pi Tour that guides the user through activating the Segment tool, holding the Control key, and clicking and dragging to create a line segment that wraps more than halfway around the sphere.

6. Add a Circle Tour that guides the user through activating the Circle tool and selecting two locations on the sphere canvas to create a circle, with an alternate method demonstrating creation by clicking and dragging.
7. Move an Object Tour that guides the user through creating a geometric object, activating the Move Objects tool, and clicking and dragging an existing object to move it.
8. Measure Line Segment Length Tour that guides the user through creating a line segment, activating the Line Segment Length tool, and selecting the segment to display its length on the sphere canvas and in the Objects panel.
9. Change Measurement Units Tour that guides the user through locating a line segment measurement in the Objects panel and changing its units between decimals, multiples of pi, and degrees.
10. Create a Right Triangle Tour that guides the user through creating a line segment, constructing a perpendicular line, creating the remaining segments of the triangle, and hiding the perpendicular line to produce a right triangle composed of three line segments.
11. Create an Equilateral Triangle Tour that guides the user through creating a line segment, constructing two circles from its endpoints, and creating two additional line segments using the circles' intersection point to form an equilateral triangle.
12. Tour popups that provide instructions and direct users toward the relevant interface elements.
13. Tour progression based on the user's actions rather than requiring a traditional "Next" button.
14. A mechanism for creating and defining new tours without manually writing each tour script.
15. Documentation describing how the tour system works and how additional tours can be created.
16. Support for opening a tour from a specific location or feature within the application.
17. Consideration for internationalization by using translation keys and the application's existing translation system.
18. Potential tour recording functionality that could allow a user to create a tutorial by demonstrating actions within Spherical Easel.
19. Potentially storing and sharing created tours using Firebase.
20. Potential improvements to the visual design and usability of tour popups.

Anticipated Platform / Tooling

- VueJS: Existing framework used to develop the Spherical Easel web application.
- TypeScript: Programming language used by the existing application and for implementing the tour system.
- Vuetify: UI framework used for the application's tooltips and tour popups.
- UITour.vue: Existing component that will serve as the starting point for implementing the tour system.
- Google Firebase: Existing service used for user accounts and cloud sharing; may also be used for storing and sharing created tours.
- Translation System: Existing translation function (T) and translation keys will be used for tour messages and internationalization.

Relevant Ethical Principles

1.01. Accept full responsibility for their own work.

3.02. Ensure proper and achievable goals and objectives for any project on which they work or propose.

3.10. Ensure adequate testing, debugging, and review of software and related documents on which they work.

3.11. Ensure adequate documentation, including significant problems discovered and solutions adopted, for any project on which they work.

7.03. Credit fully the work of others and refrain from taking undue credit.

7.05. Give a fair hearing to the opinions, concerns, or complaints of a colleague.

8.02. Improve their ability to create safe, reliable, and useful quality software at reasonable cost and within a reasonable time.

8.03. Improve their ability to produce accurate, informative, and well-written documentation.

8.04. Improve their understanding of the software and related documents on which they work and of the environment in which they will be used.